

OPERATIONS MANAGEMENT (OM)

19-205-0603 Operations Management –*syllabus*

Module I

- Operations Management-Characteristics and functions, Production process- nature and importance of production function, relationship with other business activities.
- Network techniques: Basic concept of network construction, information requirement, critical path, various slacks and floats, CPM-crashing, PERT- multi-time estimate.
- Forecasting: Importance in operations management, methods of forecasting - time series, moving average, exponential smoothening.

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- **Module II**

- Production planning and control: Scopes, objectives and functions of production planning and control.
- Product Design- Process and Functions, guiding principles, consumption cycle, product life cycle, factors affecting product design, simplification, standardization, specialization, inter-changeability.
- Inventory control: Structure of inventory problems, relevant cost, basic EOQ models, stores ledger, materials requisition sheet, materials return note, material transfer note, bin cards, just in time and lean management.

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Module- III

- Aggregate Planning: Role and need of aggregate planning, graphical and reaction rate methods of aggregate planning.
- Scheduling: Definition and scheduling decisions, Gantt charts, indexing methods, critical ratio method of loading & scheduling.
- Sequencing : Basic concepts and importance of sequencing, one machine n jobs, 2 machine n jobs, m machine n jobs problems.

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• **Module IV**

- Plant Location: Factors influencing location, significance of sites in Urban, Semi-urban and Rural areas.
- Plant Layout: Types, need for layout, layout design process- factors, determination of equipment and employee requirement, production rate determination, space determination, block plan, systematic layout planning.
- Material handling: The principles of materials handling, classification of equipment's and its selection factors.
- Maintenance & Replacement: Different types of maintenance, merits and demerits, operational and economic aspects.
- Replacement of equipment, reasons and basic methods, concept of depreciation.

Operations Management (OM)

- Operations management (OM) is the administration of business practices to create the highest level of efficiency possible within an organization.
- It is the process of planning, organizing and regulating the operations or activities.
- It is concerned with converting materials and labor into goods and services as efficiently as possible to maximize the profit of an organization

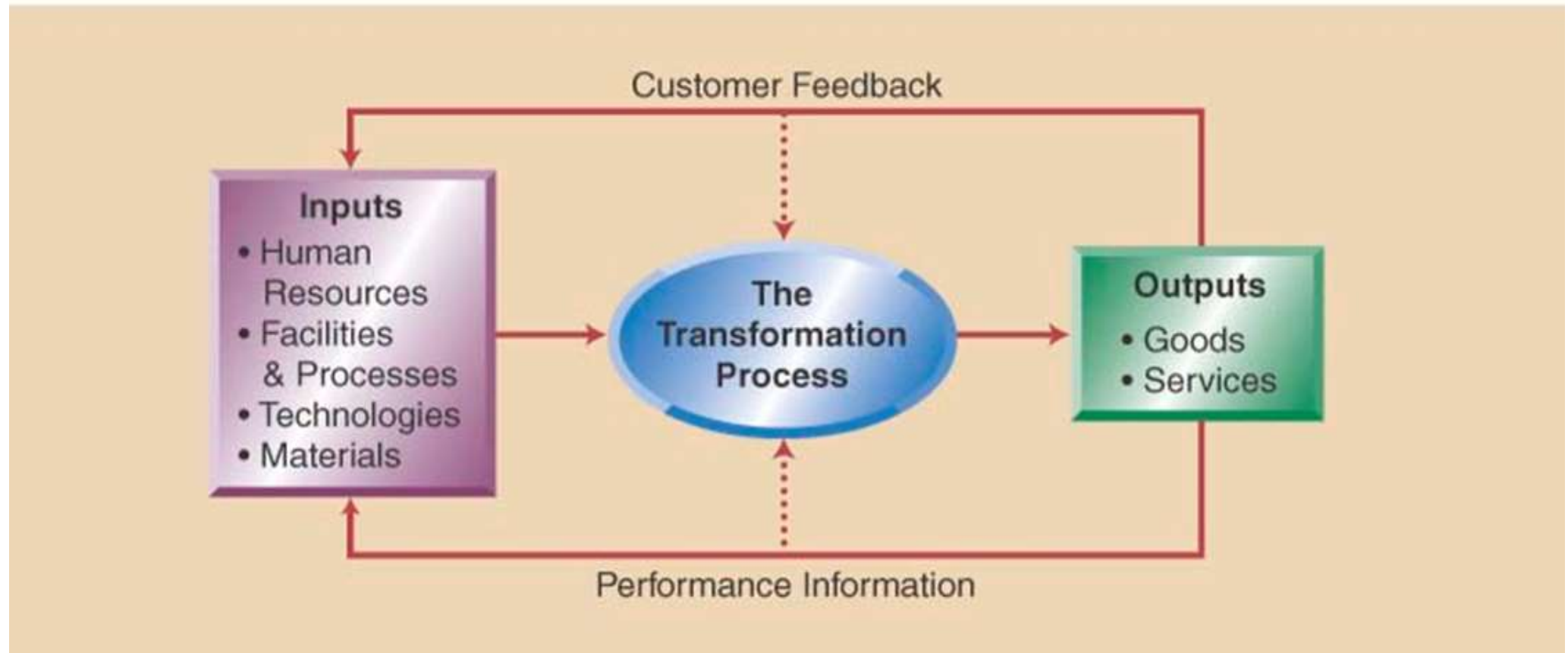
OM

- Operations management handles various strategic issues, including determining the size of manufacturing plants and project management methods and implementing the structure of information technology networks.
- Other operational issues include the management of inventory levels, including work-in-process levels and raw materials acquisition, quality control, materials handling, and maintenance policies.
- *The business function responsible for **planning, coordinating, and controlling** the **resources** needed to produce products and/or services for an organization.*

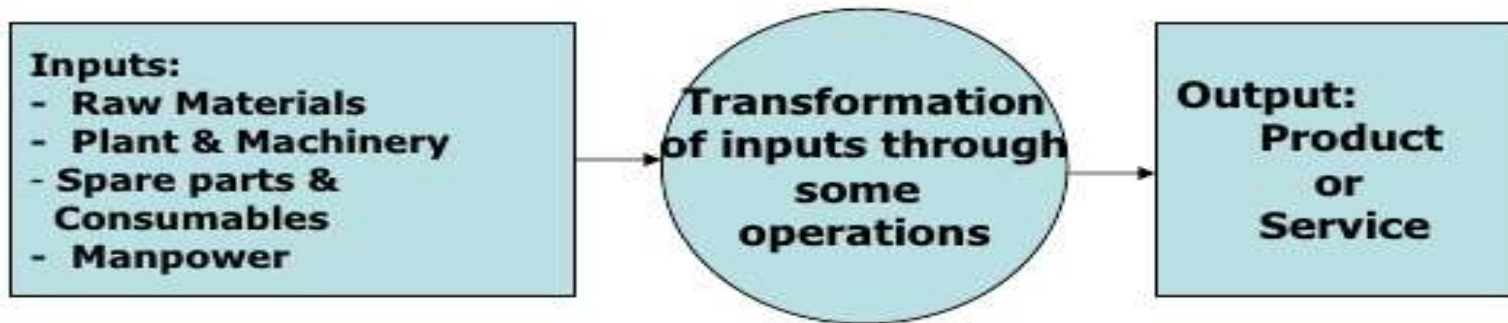
Operations management

- The important responsibility of operations manager is about **processing inputs and make outputs** by **controlling** required parameters.
- The primary keywords are;
 - *Resources
 - *Systems and transformation
 - *Value addition activities.

OM-Transformation process

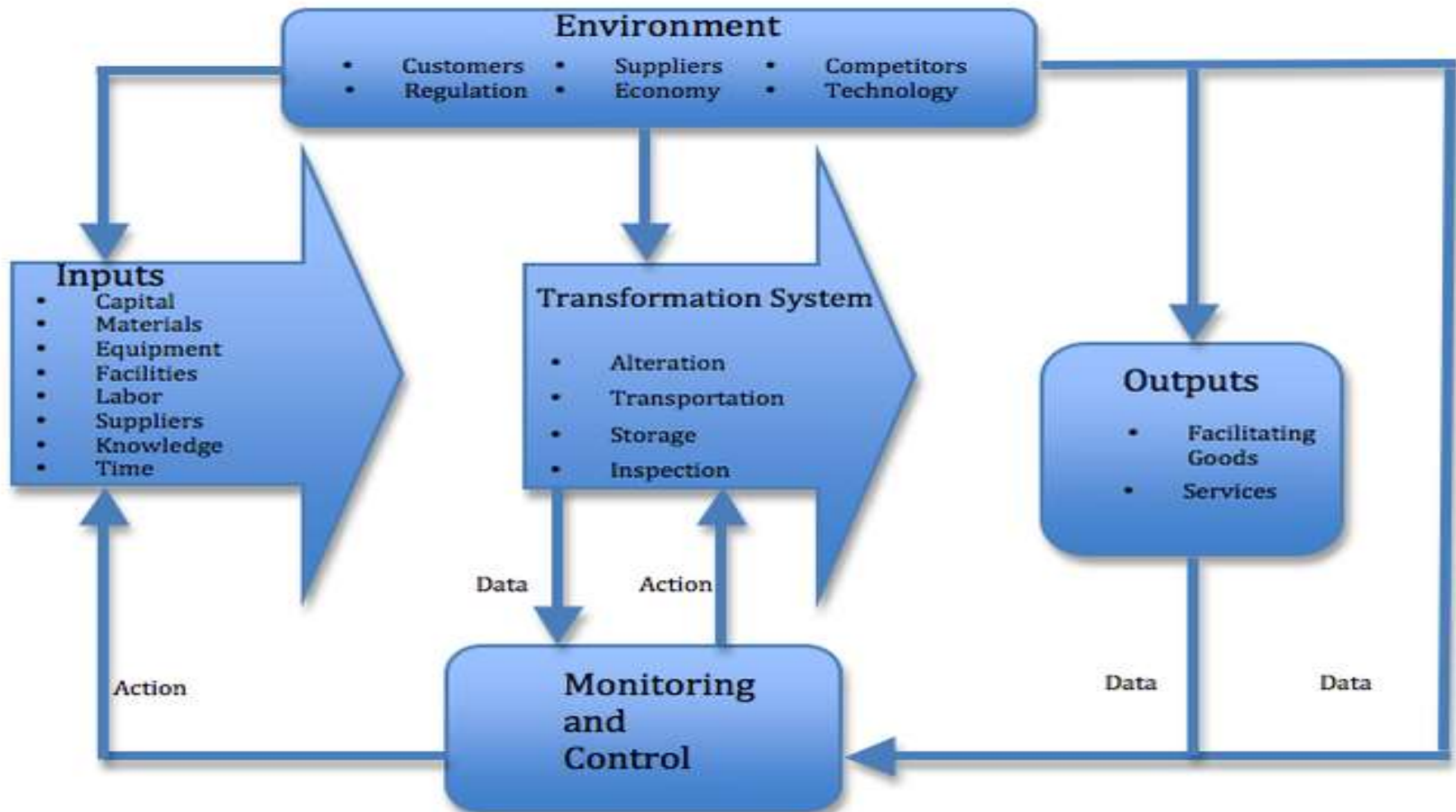


OPERATION MANAGEMENT



Example of Cement Plant





OM-Objectives

Objectives are classified in to;

- Performance objectives
- Cost objectives

Performance Objectives

- Productivity-efficiency
- Effectiveness
- Quality
- Lead time-time elapsed in conversion process
- Capacity utilization
- Flexibility
- *Right quality, right quantity, right time and right price*

Cost objectives

Visible(explicit) cost

- Material cost
- Direct and indirect labour
- Scrap/rework cost
- Maintenance cost

Cost objectives-Implicit (invisible/hidden) cost

- Inventory carrying cost
- Lost sales, stockouts, shortages
- Delayed deliveries
- Material handling
- Inspection
- Grievances, dissatisfaction
- Downtime

Functions and Roles

- Managing transportation and distribution
- Managing and maintaining facilities
- Conducting enterprise resource planning (ERP)
- Forecasting for planning
- Capacity planning
- Navigating industrial labor relations
- Analyzing the value chain
- Optimizing resource allocation
- Eliminating waste and bottlenecks
- Continuously improving processes
- Executing a company's strategic plan

SCOPE OF OPERATION MANAGEMENT

- Operations Management concerns with the conversion of inputs into outputs, using physical resources, so as to provide the desired utilities to the customer while meeting the other organizational objectives of effectiveness, efficiency and adoptability.
- It distinguishes itself from other functions such as personnel, marketing, finance, etc. by its primary concern for ‘conversion by using physical resources’.
- Following are the activities, which are listed under Production and Operations Management functions:
 1. Location of facilities.
 2. Plant layouts and Material Handling.
 3. Product Design.

SCOPE OF OPERATION MANAGEMENT

- 4. Process Design.
- 5. Production Planning and Control.
- 6. Quality Control.
- 7. Materials Management.
- 8. Maintenance Management

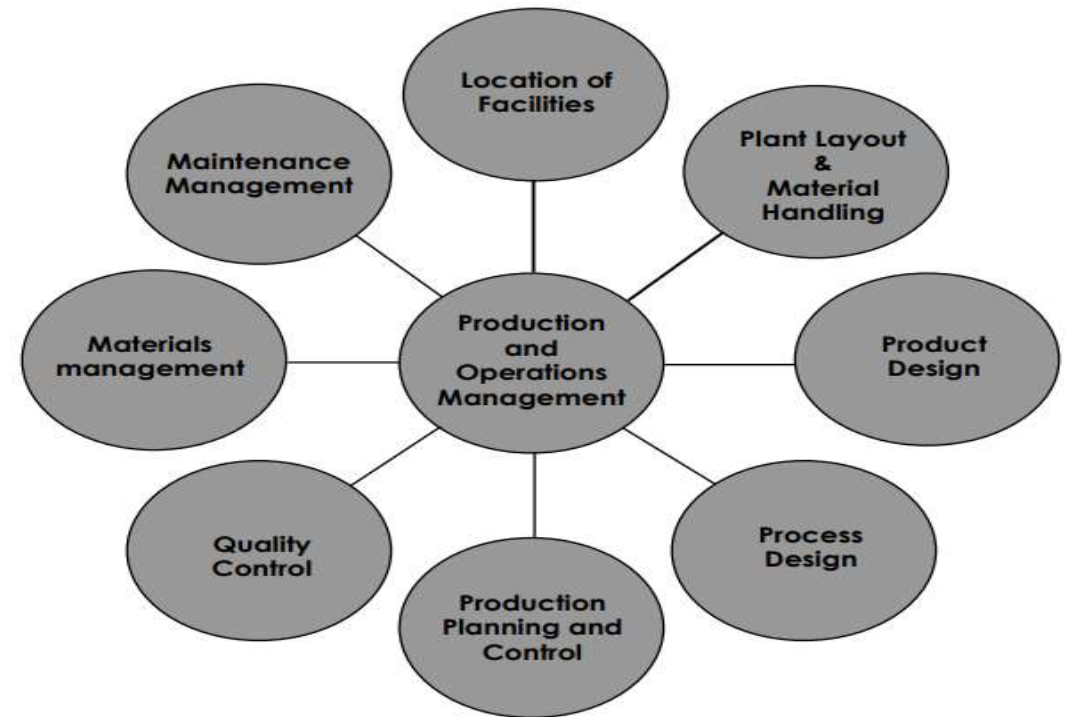


Fig. 1 : Scope of production and operations management

Relationship of Operations Management to Other Departments

- Operations managers facilitate cooperation between departments and ensure that department and company goals are aligned, and to standardize logistical approaches to budget and project constraints. Operations teams must collaborate with all other departments to be effective.



Relationship of Operations Management to Other Departments

- Operations managers are deeply involved in strategy, in addition to their daily production roles. From designing and testing processes to facilitating interdepartmental collaboration, we have outlined some key strategy and tactics points below:
- **Data Use:** Analytics are essential for strong planning, adjustments, and decision making. Two common types are efficiency metrics and effectiveness metrics.
- **Inventory Analysis:** To manage inventory in the supply chain, ABC analysis (also called Pareto analysis) comes into play. This method divides inventory into three categories: A, B, and C. Category A has the highest value and tightest controls, and Category C has the lowest value and loosest controls.

Relationship of Operations Management to Other Departments

- **Data Challenges:** Data is often siloed, which makes it difficult to compare. But newer systems and setups make this process easier and help analysts and managers to examine data in new, helpful ways.
- **Process Design:** Researching, forecasting, and developing a sound process takes expertise and energy, but the results can be lasting.
- **Forecasting and Goal Setting:** The best forecasting often combines a look at historical data with an analysis of changing conditions.

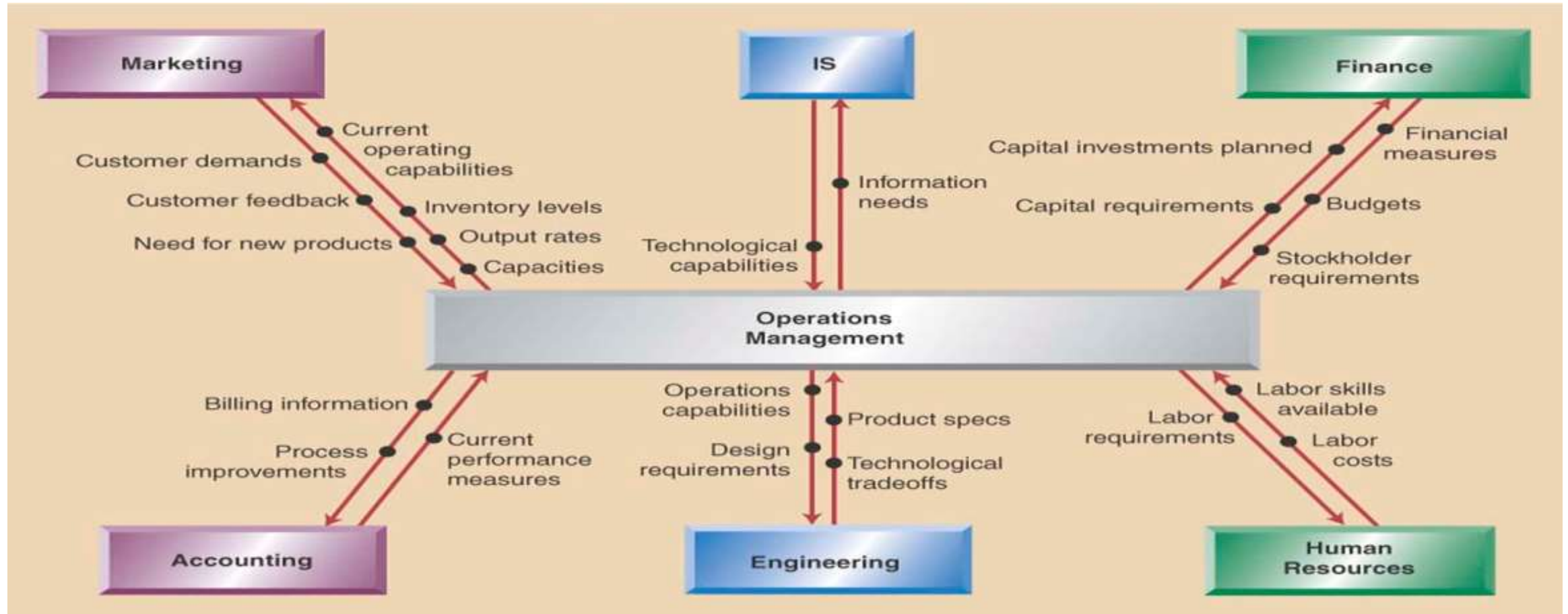
Relationship of Operations Management to Other Departments

- **Collaboration Among Departments:** With good communication and collaboration, operations management can work effectively with finance, sales, marketing, human resources, and other departments.
- **Being Green:** Ecological soundness has become a strategic and legal necessity at companies nowadays, especially in manufacturing.
- **Managing People:** With all the advancements in machinery and technology, people remain critical to the equation, though often in different types of jobs.

Production system-Inputs (6M's)

- Men
- Machine
- Material
- Money
- Method
- Management

Business Information Flow



NETWORK ANALYSIS

To Plan, monitor and control process/measures for projects

Project

- Project- defined as a unique activity which has a definite beginning and definite end, consumes time and resources.
- Project- is a set of inter related activities organized for a common goal.
- Project is any human undertaking with a clear beginning and clear ending.
- Examples: ?

Network

- Interrelationship of activities or events of a project with a mention on time required for the completion of each activity.
- Is a schematic representation of work to be done (activities) drawn in such a way that relationships between work items are logically defined.

Network Analysis

31.7. TERMS RELATED WITH CPM AND PERT

The CPM and PERT techniques have their own language which can be understood only when the following terms and their representation is known.

Event :

An event is a specific instant of time which indicates the beginning or end of the activity. Event is also known as a junction. It is represented by a circle and the event number is written within the circle. Unlike an activity, an event consumes neither time nor resources.

Activity :

Every project consists of number of job operations or tasks which are called activities. An activity is any time or resource consuming part of the project which has definable start and finish. It may be a process such as moulding, finishing, cutting etc.; a material handling or material procurement cycle.

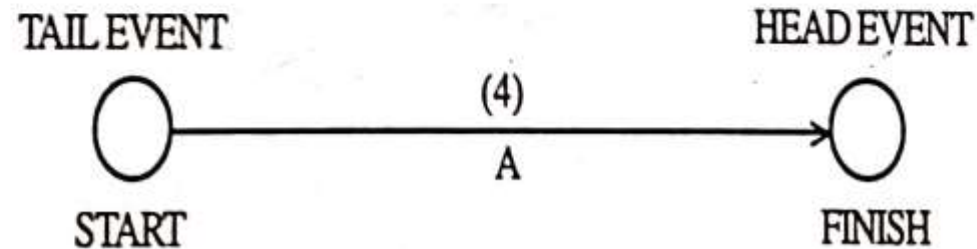


Fig. 31.11. Activity.

Network Analysis

An activity is represented by an arrow in the network diagram. It begins in start event which is also known as 'Tail Event', and ends in completion event or 'Head Event'. An activity is normally given a name like A, B, C etc. which is marked below the arrow and the estimated time to accomplish the activity is marked above the arrow.

Following example clarifies the difference between an event and activity.

- | | |
|------------------------------------|---------------|
| Start machine installation | - An event |
| Machine installation | - An activity |
| Completion of machine installation | - An event. |

Classification of Activities :

Activities are classified as :

- (a) Critical activities
- (b) Non-critical activities
- (c) Dummy activities.

(a) **Critical Activities.** In a network diagram critical activities are those which if consume more than

Network Analysis

their estimated time, the project will be delayed. More attention should be paid to the critical activities. The expected completion dates cannot be met, even if one of the critical activities is delayed.

A critical activity is marked either by a thick arrow or a double line arrow to distinguish it from non-critical activities.

(b) **Non-critical Activities.** Such activities have a provision of float or slack so that, even if they consume a specified time over and above the estimated time, the project will not be delayed.

(c) **Dummy Activities.** When two activities start at the same instant of time like A and B, the head events are joined by dotted arrows and this is known as dummy activity.

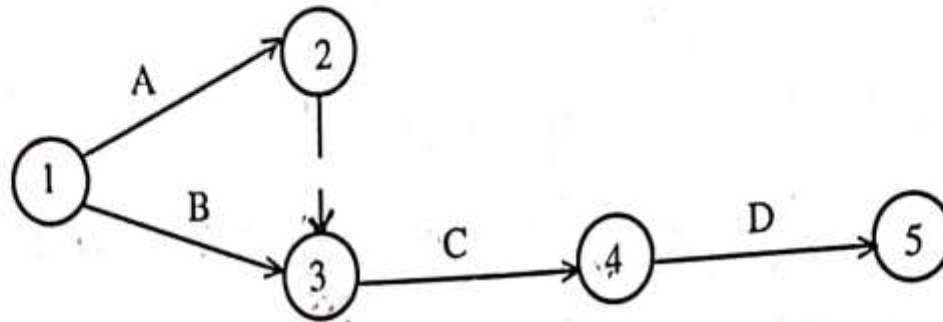
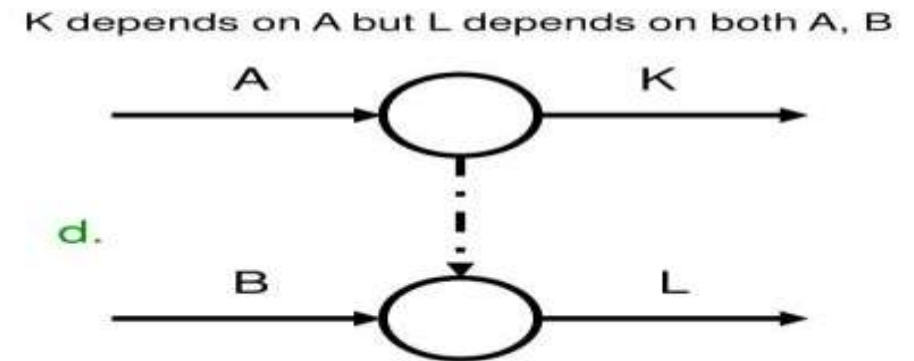
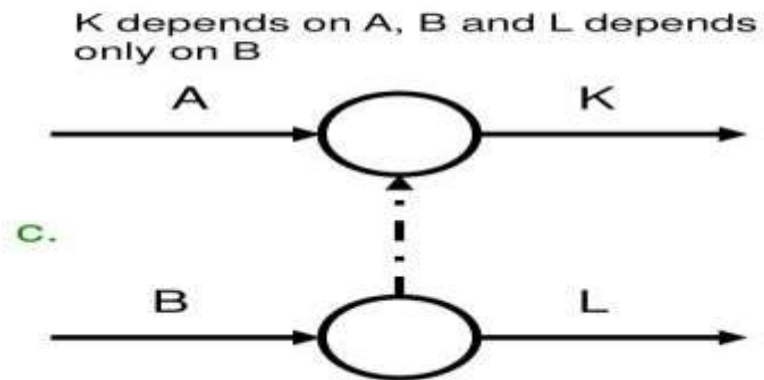
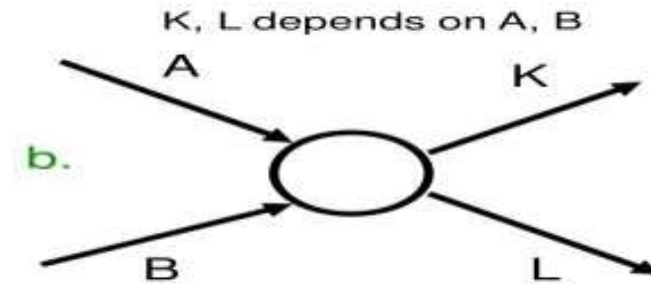
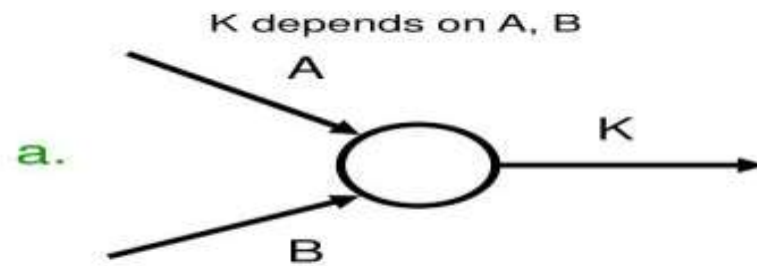


Fig. 31.12. Dummy Activity.

The dummy activity is included in the network to show that the activity C cannot be started until activities A and B are completed. The dummy activities however complete the logical concept, but do not consume any time or resources. A dummy activity is thus a connecting link for control purpose or for maintaining uniqueness of activity.

Activity

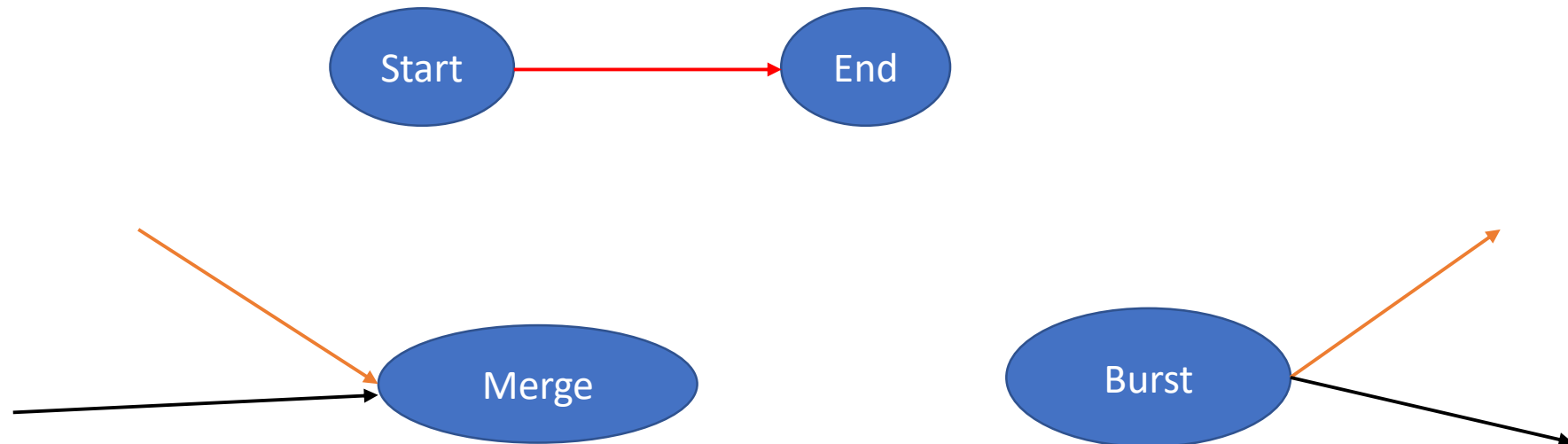
Activity on Arrow (AoA) Network Diagrams



Activity

- Predecessor
- Successor
- Critical
- Dummy

Events

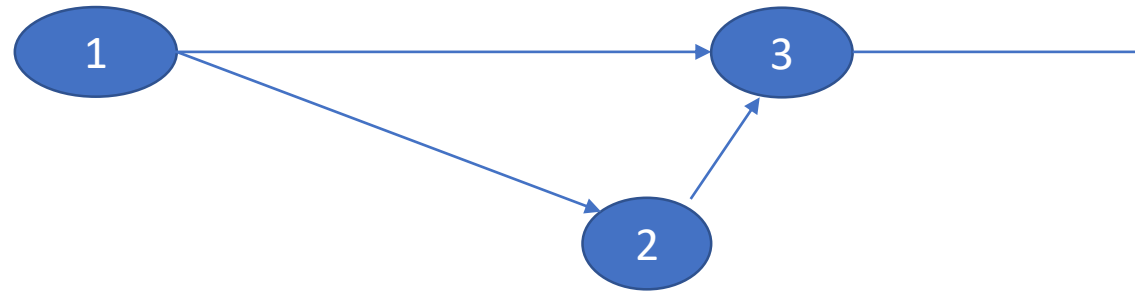


Dummy activity

- Wrong



- Correct



Events

- Start event
- End event
- Merge event
- Burst event

Network Analysis

Network diagram or Arrow diagram :

It is a pictorial representation of a project plan showing the inter-relationship and interdependencies between the various activities in a sequence; in which they are to be performed to complete the project. Fig. 31.13 shows a simple network consisting of 5 events and its activities to be performed are designated as A to F. Event 1 is the start of the project and event 5 is the completion of the project.

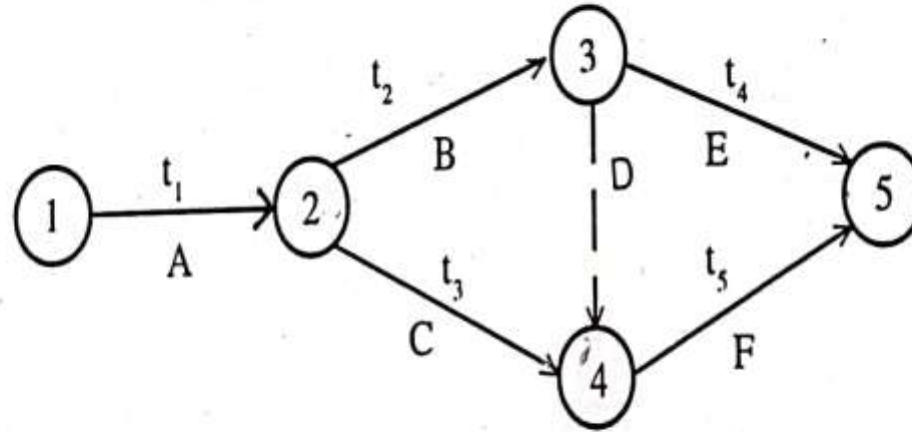


Fig. 31.13. Network Diagram.

The logic of the network is that activities B and C cannot start until event 2 is reached or in other words until activity A is completed. Activities B and C can be done concurrently. As shown in the figure the activity D is a dummy activity connecting events 3 and 4 and hence it is shown by dotted line. This dummy activity denotes that activity F cannot start unless both activities B and C are completed.

Network Analysis

31.8. RULES OR NETWORK CONSTRUCTION

1. The length of the arrow does not indicate the time which the activity takes or the resources which the activity consumes.
2. The direction of the arrow indicates the direction of work flow i.e. it shows the dependencies and the sequence in which the activities are to be performed.
3. No activity can be shown twice in the network unless it occurs twice in the project. Normally in such cases it will bear a different activity identification.
4. No two events will be directly connected by more than one activity.
5. An event cannot occur until all activities leading to it are completed and
6. No activity can begin until its immediate preceding event has occurred.

Network Analysis

Pre-operations. Pre-operations are the operations which precede the operation under consideration e.g. machine installation can only be done after machine foundation is completed. In this case, machine foundation is the pre-operation of machine installation.

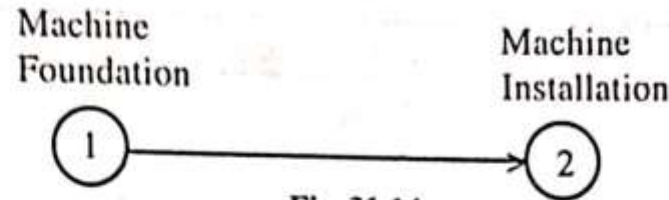


Fig. 31.14.

The arrow connecting these two circles shows the reaction between them and indicates that the operation on the left end of an arrow should be completed before the beginning of the operation on the right end of arrow.

Sometimes, an operation has more than one pre-operation. For example, roofing of a workshop shed can only be started when superstructure and the steel fabrication is completed. Thus steel fabrication and superstructure are the preoperations of the roofing as shown in Fig. 31.15.

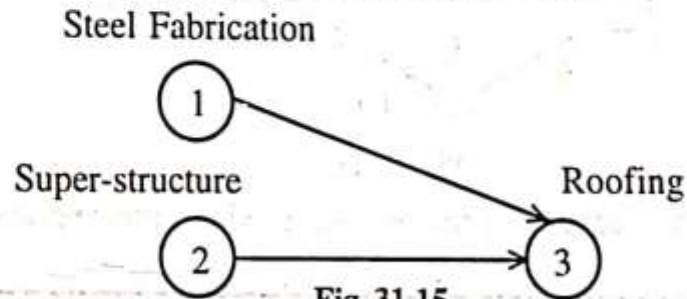


Fig. 31.15.

Post-operations. Post-operations are the operations which succeed the operation under consideration. In the previous example, machine installation is the post-operation of the machine foundation.

Concurrent Operations. These are the operations which can be started simultaneously (at the same time).

Network Analysis

Earliest Start Time (EST). It is the earliest possible time at which an activity can start, and is calculated by moving from first to last event in the network diagram.

Earliest Finish Time (EFT). It is the earliest possible time at which an activity can finish.

$EFT = EST + \text{duration of that activity.}$

Latest Start Time (LST). It is the latest possible time by which an activity can start without delaying the date of completion of the project.

$LST = LFT - \text{duration of that activity.}$

Latest Finish Time (LFT). It is the latest time by which the activity must be completed so that the scheduled date for the completion of the project may not be delayed. It is calculated by moving backwards i.e. from last event to first event of the network diagram.

Float or Slack. Float or Slack means a margin of extra time over and above its duration which a non-critical activity can consume without delaying the date of completion of the project.

Float or Sack = Time available for completion of the activity – Time necessary to complete the same.

Slack is used with PERT and float with CPM but in general practice they may be used interchangeably.

Float (LST – EST) or (LFT – EFT)

Network Analysis

Critical Path :

A path along the network in which Earliest Finish Time and Latest Finish Time are equal is known as critical path. It is that sequence of activities which decides the total project duration. It is the longest path and consumes maximum time. Critical path is formed by critical activities and has zero float. It is the path which the management is most anxious to determine, shorten and monitor. If the project has to be completed in lesser time, then the duration of the activities lying in the critical path have to be shortened. The critical path is represented by double or thick arrow line to distinguish it clearly.

Project network -Methods

- CPM- Critical Path Method
- PERT-Programme Evaluation and Review Technique

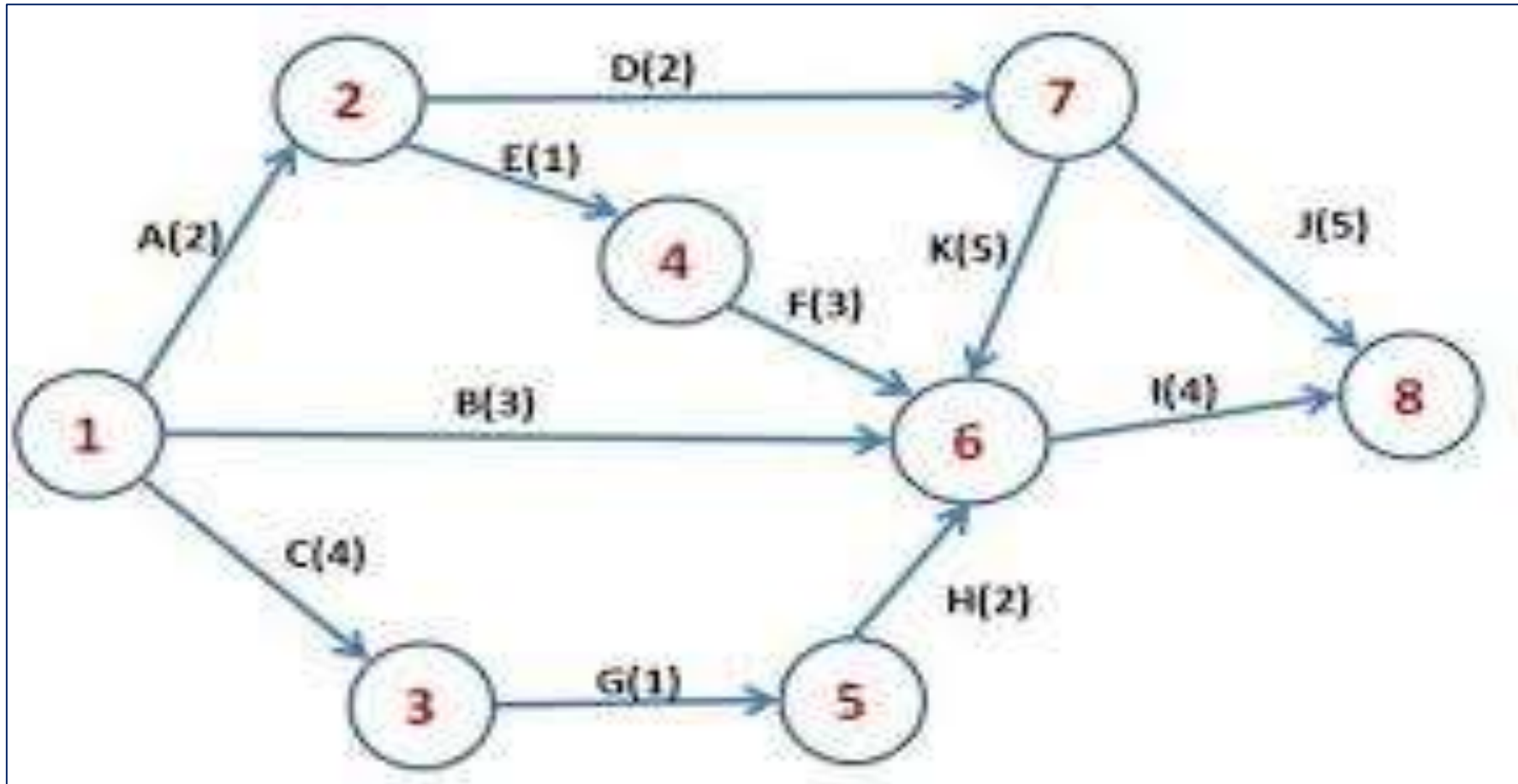
CPM

- Activity oriented
- Deterministic model
- Single time estimate possible
- Best suited for construction

PERT

- Event oriented
- Probabilistic model
- 3 time estimate (T_p , T_o and T_m)
- Expected time (T_e) = $(T_p + 4T_m + T_o)/6$
- Research and development projects

Project Network



Network paths

Path (activity based)B-I	Path (event based)	Duration
A-D-J	1-2-7-8	9
A-E-F-I	1-2-4-6-8	10
B-I	1-6-8	7
C-G-H-I	1-3-5-6-8	11
A_D_K_I	1-2-7-6-8	13

Critical path

- Path having maximum duration
- This duration is the minimum time required for the completion of project.
- Activities in the critical path are the critical activities.
- This path is represented by a double line.

problem

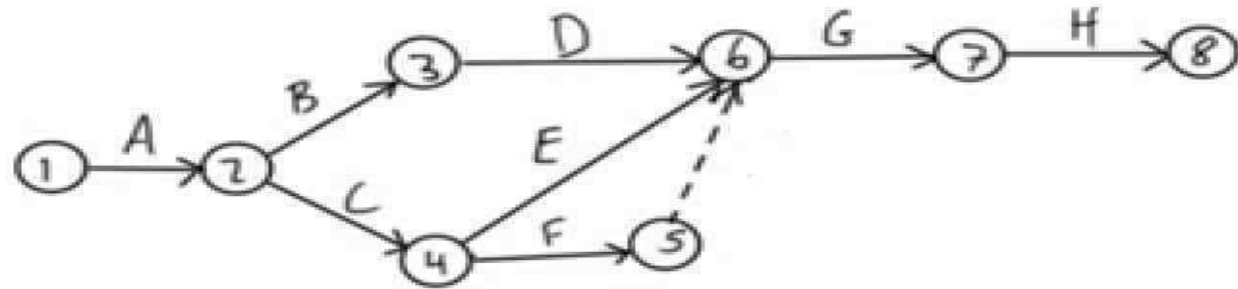
ACTIVITY	IMMEDIATE PREDECESSOR	ACTIVITY DURATION (DAYS)
A	-	3
B	A	5
C	A	4
D	B	3
E	C	5
F	C	2
G	D,E,F	6
H	G	1

CPM-NETWORK-2

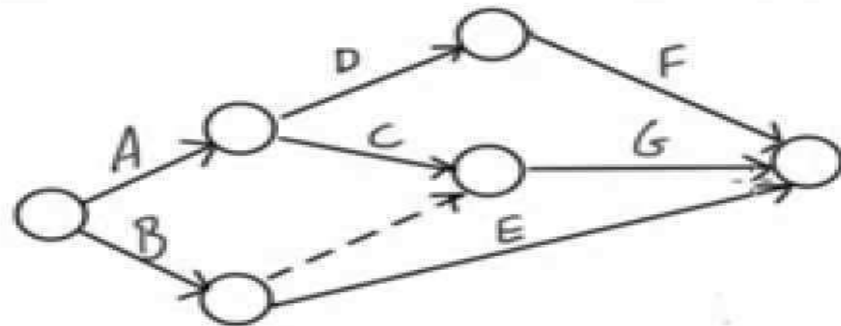
ACTIVITY	IMMEDIATE PREDECESSOR	ACTIVITY DURATION (DAYS)
A	-	5
B	-	4
C	A	3
D	A	6
E	B	3
F	D	4
G	B,C	4

Problems:

Activity	Predecessor
A	-
B	A
C	A
D	B
E	C
F	C
G	D,E,F
H	G



Activity	Predecessor
A	-
B	-
C	A
D	A
E	B
F	D
G	B,C



Problem-1

Example A small scale industrial unit consists of 6 activities as given below :

<i>ACTIVITY</i>	<i>TIME IN DAYS</i>	<i>PRE-OPERATION</i>
A	5	None
B	6	A
C	5	B
D	4	A
E	3	D
F	4	C, E

Draw the network diagram and calculate EST, LST, EFT, LFT and floats, mark the critical path and find total project duration.

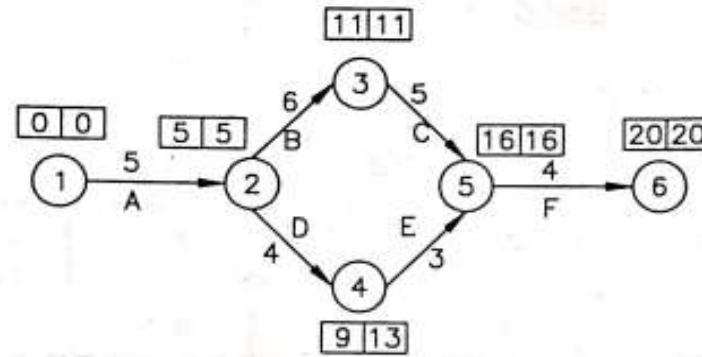


Fig. 1. Network Diagram.

Activity 1	Duration 2	EST 3	LST 4	EFT 5	LFT 6	Total 7	Remarks 8
1-2 (A)	5	0	0	5	5	0	Critical
2-3 (B)	6	5	5	11	11	0	Critical
3-5 (C)	5	11	11	16	16	0	Critical
2-4 (D)	4	5	9	9	13	4	Non-Critical
4-5 (E)	3	9	13	12	16	4	Non-Critical
5-6 (F)	4	16	16	20	20	0	Critical

The critical path is 1-2-3-4-5-6 as shown in the network diagram. It is seen that for activities on the critical path EST for the succeeding activity = LFT for the proceeding activity. The total project duration is 20 days.

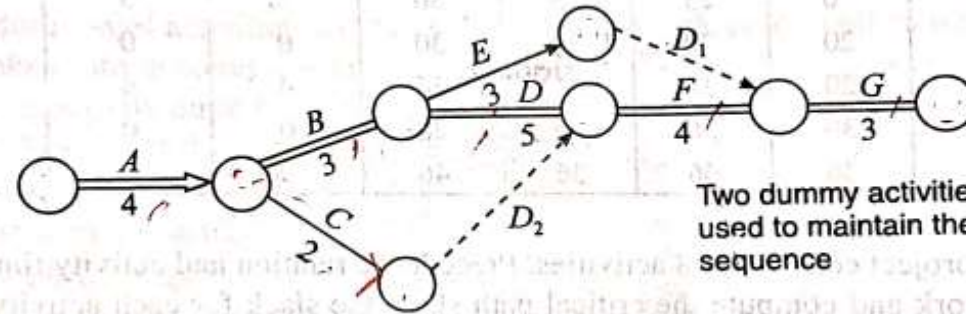
Problem-2

Problem : The activity details and their predecessors are given below along with their activity times.

Construct the network diagram

Activity	Predecessors	Activity time (Weeks)
A	—	4
B	A	3
C	A	2
D	B	5
E	B	3
F	C, D	4
G	E, F	3

Solution: The network diagram is shown in the Fig. 20.6.



Two dummy activities are used to maintain the logical sequence

Fig. 20.6 : Network diagram.

The slack for the activities is calculated as shown in Table 20.1.

Table 20.1 : Computation of slack.

Activity	Activity Time	ES	EF	LS	LFT	Slack
A	4	0	4	0	4	0
B	3	4	7	4	7	0
C	2	4	6	8	12	2
D	5	7	12	7	12	0
E	3	7	10	9	12	2

Activity	Activity Time	ES	EF	LS	LFT	Slack
F	4	12	16	12	16	0
G	3	16	19	16	19	0

The critical path is one that connects the activities with zero slack. The critical paths

A — B — D — F — G

Problem-3

Problem : The activities involved in a small project are given below along with relevant information. Construct the network and find the critical path

Find the floats for each activity.

Activity	1-2	1-3	2-3	2-4	3-4	4-5
Duration	20	25	10	12	6	10

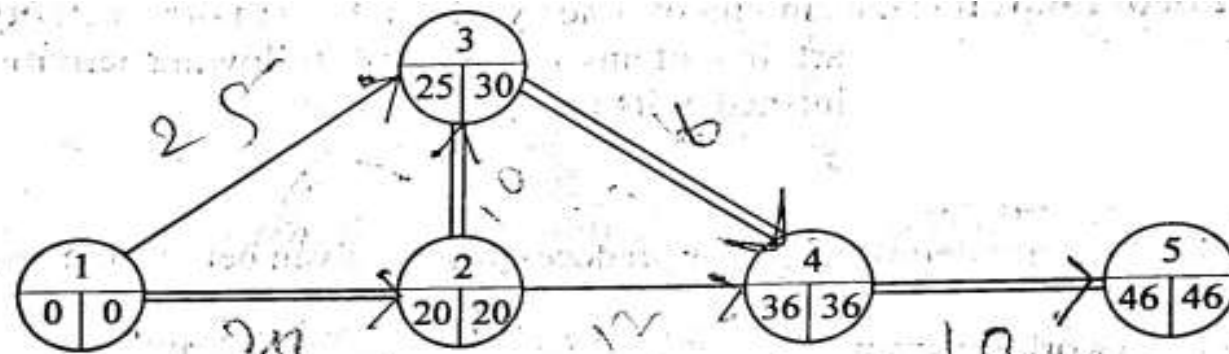


Fig. : Network diagram.

Critical path is 1 – 2 – 3 – 4 – 5
 The floats are calculated as shown in Table

Table : Calculation of floats

Activity	Duration	Earliest		Latest		Float		
		Start	Finish	Start	Finish	Total	Free	Independent
1 – 2	20	0	20	0	20	0	0	0
1 – 3	25	0	25	5	30	5	5	5
2 – 3	10	20	30	20	30	0	0	0
2 – 4	12	20	32	24	36	4	4	4
3 – 4	6	30	36	30	36	0	0	0
4 – 5	10	36	46	36	46	0	0	0

Problem-4

Example 1. Calculate EST and LFT, total float and project duration for the network given in Fig. 2.1.2 below, also calculate the total project duration and show the critical path on the network.

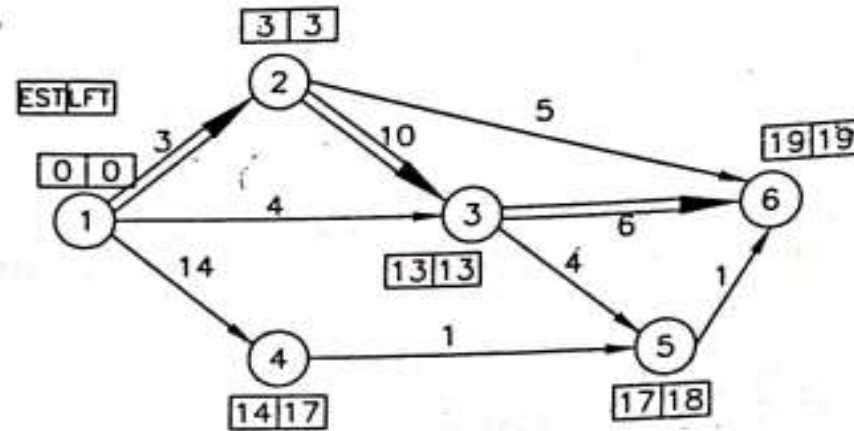


Fig. 2.1.2

Solution. The EST and LFT are as shown in Fig.

<i>Activity</i>	<i>Duration</i>	<i>EST</i>	<i>LST</i>	<i>EFT</i>	<i>LFT</i>	<i>Total Float</i>	<i>Remarks</i>
<i>1</i>	<i>2 (days)</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>
1-2	3	0	0	3	3	0	Critical
1-3	4	0	9	4	13	9	Non-critical
1-4	14	0	3	14	17	3	Non-critical
2-3	10	3	3	13	13	0	Critical
2-6	5	3	14	18	19	11	Non-critical
3-5	4	13	14	17	18	1	-do-
3-6	6	13	13	19	19	0	Critical
4-5	1	14	17	15	18	3	Non-critical
5-6	1	17	18	18	19	1	-do-

Total project duration 19 days. Critical path is 1-2-3-6.

Problem 5

Problem. A project consists of 8 activities. Precedence relation and activity times are given. Draw the network and compute the critical path show the slack for each activity in a tabular form.

Activity	Immediate Predecessor	Activity Time (Weeks)
P	-	12
Q	-	20
R	-	28
S	R	12
T	P, Q	28
U	T, S	12
V	S	8
W	U, V	8

Problem-6

Example 7. Draw the network diagram for the following logic :

- (i) A is the first operation on the project.
- (ii) B & C can be done concurrently and both must follow A.
- (iii) B must precede A.
- (iv) E cannot begin until both B & C are completed.
- (v) F is dependent on the completion of D and E.
- (vi) F is the last operation on the project.

CRASHING (TIME COST TRADE OFF)

- The process of reducing project duration is called crashing
- Crashing will be done at the cost of extra resources....in critical path.
- Any reduction in project time means reduction in indirect costs.
- Direct costs?
- Indirect costs?

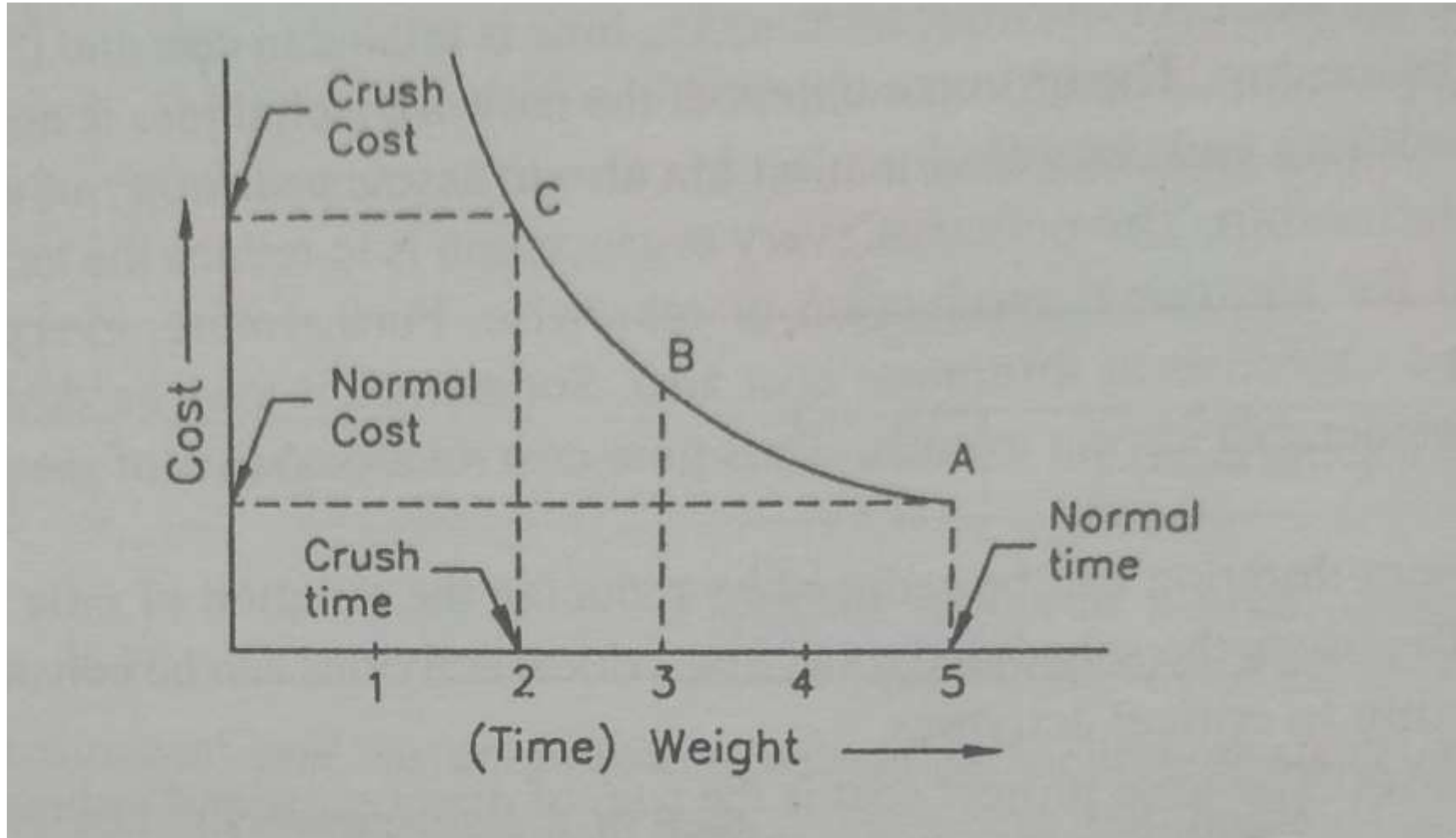
Direct and Indirect Costs

- Direct costs: material cost, machinery used in the project, labour-both direct and subcontracting.
- Indirect costs: Overheads, office and administrative expenses, penalty cost. This varies with time

Crashing Of Network(Time-Cost Relationship)

- In **CPM/PERT** techniques, Time is related to cost and the objective is to develop an optimum time-cost relationship.
- The policy of every organization is to reduce the target time so that the time saved can be utilized for additional production or otherwise
- Furthermore every organization wants to accomplish the desired objectives at minimum cost also.

Crash Point and Normal Point



Crash Point and Normal Point

- **Normal Time:** It is the standard time associated with normal resource of an organization to perform the activity.
- **Crash Time:** It is the minimum possible time in which an activity can be completed by employing extra resources. It is the time beyond which the activity cannot be shortened by any amount of increase in resources.
- **Normal Cost:** Expenditure incurred on Normal resource for completing any activity in normal time.
- **Crash Cost:** The total expenditure incurred on normal and additional resources for crashing the time.

Crash Point and Normal Point

The time-cost relationship is not linear, but the relationship can be approximated by a straight line which is called as the 'cost-slope'.

The cost slope formula can be represented as :

$$\text{Cost slope} = \frac{\text{Crashed cost} - \text{Normal cost}}{\text{Normal time} - \text{Crash time}}$$

Cost slope represents the extra cost of shortening the duration of the activity by one time unit. For reducing the activity duration the management may agree for extra expenditure but to keep this expenditure minimum we must concentrate on those activities for which cost slope is minimum.

Cost slope

- C_t - is the minimum time possible for the activity
- N_t - Time for the activity at minimum cost
- Costs associated with these times are called Crash cost (C_c) and Normal cost (N_c)
- Cost slope = $(C_c - N_c) / (N_t - C_t)$

Crashing procedure

1. Construct the network diagram and identify the critical path
2. Calculate the cost slope for different activities
3. Crashing the activities in critical path as per ranking(first crash the activity with lowest cost slope)
4. Determine the total cost of the project after every crashing stages.
5. This includes both direct and indirect costs.
6. Finally find the total cost and duration after crashing all possible activities in the critical path.

PERT

- PERT gives a way of identifying critical activities right at the planning stage so that management is aware of these potentially troublesome areas and exercise proper control over them so as to complete the project within the scheduled time.
- It is developed actually as a research and development planning tool where activity timings could not be estimated with enough certainty.
- It can be employed at those places where a project cannot be easily defined in terms of time or resources required.

PERT

- Because of the uncertainty of activity timings PERT acquires the shape of probabilistic model.
- It uses linear programming and probability concepts for planning and control activities.
- Probability concepts helps in estimating activity timing and the maximization of the predetermined objectives is achieved through the linear programming.
- PERT is mainly concerned with events and it is thus event-oriented system.

PERT

- Event oriented
- Probabilistic model
- 3 time estimate (t_p , t_o and t_m)
- Expected time (t_e) = $(t_p + 4t_m + t_o)/6$
- Variance = $[(t_p - t_o)/6]^2$
- Research and development projects

Estimation of Activity Time

31.10. ESTIMATION OF ACTIVITY TIME

PERT is based on the assumption that operation time cannot be estimated accurately because of the lack of past data and uncertainties involved in working. To take these uncertainties into account, PERT uses three time estimates for each activity.

1. **Optimistic time (t_o)**. It is the shortest time in which an activity can be completed assuming that everything goes exceptionally well. It has low probability of occurrence.

2. **Most likely time (t_m)**. It is the most likely time required to complete the activity taking into consideration all favourable and unfavourable elements. This estimate of time lies between the optimistic and pessimistic time.

3. **Pessimistic time (t_p)**. It is the time which an activity will take to complete if everything turns out to be against expectation (*i.e.* under adverse conditions). Similar to optimistic time it has low probability of occurrence.

Estimation of Activity Time

If the probability distribution of these times estimates is drawn, the curve would approximately follow (beta) distributions.

Expected Time (t_e). The three time estimate t_o , t_m , t_p are combined statistically to develop the expected time t_e . It is assumed that t_p and t_o are about equally likely to occur whereas the probability of occurrence of t_m is 4 times that of t_o and t_p .

$$\text{Therefore, } t_e = \frac{t_o + 4t_m + t_p}{6}$$

In order to maintain the reliability in the time duration (t_e), the variability in the time estimates should be measured and this in turn can be done by standard deviation or variance of their probability distribution. PERT simplifies the calculation of standard deviation of the time.

$$\text{Estimates, } \sigma t = \frac{(t_p - t_o)}{6} \text{ and variance, } V_i = \sigma t^2$$

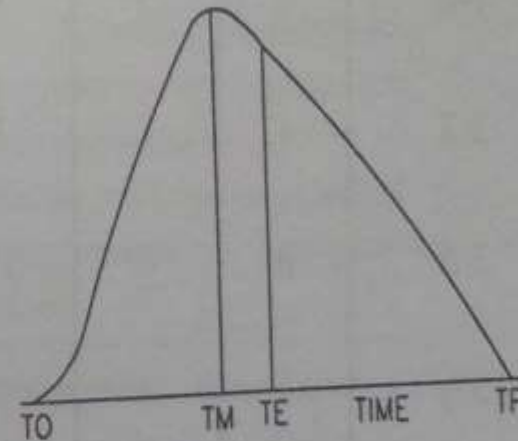


Fig. 31.25. Distribution.

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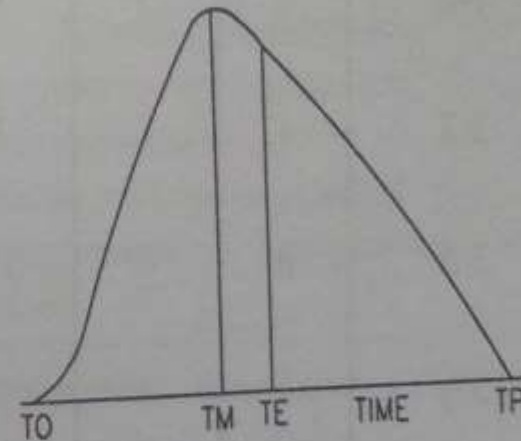


Fig. 31.25. Distribution.

Difference Between PERT And CPM

31.11. DIFFERENCE BETWEEN CPM AND PERT

Though both PERT and CPM are used synonymously ; they differ from each other in the following respects :

No.	PERT	CPM
1.	PERT is used where the emphasis is on shortening project execution time without too much concern for cost implication.	CPM is used where the emphasis is on optimising resource allocation and minimizing overall cost for a given projection execution time.
2.	A probabilistic model with uncertainty in a activity duration. Three time estimates are used to make allowances for uncertainties.	A deterministic model with well known activity time based on past experience. It uses single time estimate.
3.	An event oriented technique.	CPM is activity oriented.
4.	The use of dummy activities is required for representing the proper sequencing.	The use of dummy activities is not necessary. The arrow diagram thus becomes slightly simpler.
5.	Especially suitable in defence projects, R and D where the activity times cannot be readily predicted.	Suitable for problems in industrial plant maintenance, civil construction projects etc.

However, over a period of time the difference between the two has narrowed down and there is very little difference between the two as they stand today.

Probability of completion

- Find the duration in the critical path
- Find the total variance in the critical path
- Square root of total variance will give you the standard deviation
- Compute standard normal variate $Z=(x-\mu)/\sigma$
- X is the anticipated /expected project duration under question
- μ is the duration in the critical path..mean time of project calculated
- σ is the standard deviation in the critical path.

Slack and float

- **slack** is referring to the amount of time that an activity can start later than originally planned
- **float** is about the time when an activity takes longer than originally planned. ...
- Total **float** is the amount of time that an activity can be delayed without delaying the completion of the project.

Slack Versus Float

- The terms "slack" and "float" are often used interchangeably.
- However, the main difference between float and slack is that slack is typically associated with inactivity (PERT), while float is associated with activity (CPM).
- Slack time allows an activity to start later than originally planned, while float time allows an activity to take longer than originally planned.

Slack or Float

- = Latest Start Time - Earliest Start Time

OR

Latest Finish Time - Earliest Finish Time